1.	$\int\left(\frac{8}{3}x^3-\frac{1}{2}x^{-\frac{1}{2}}-5\right)\mathrm{d}x=\frac{8}{3}\times\frac{x^4}{4}-\frac{1}{2}\times 2x^{\frac{1}{2}}-5x+c$ $=\frac{2}{3}x^4-x^{\frac{1}{2}}-5x+c$	M1 A1 A1 A1 **(4 marks)**

2 (a)	$3^{3x} = \left(3^x\right)^3 = y^3$	B1 **(1)**
(b)	$\dfrac{1}{3^{x-2}} = \dfrac{1}{3^x \times 3^{-2}} = \dfrac{9}{y}$	M1 A1 **(2)**
(c)	$\dfrac{81}{9^{2-3x}} = \dfrac{9^2}{9^{2-3x}} = 9^{2-(2-3x)} = 9^{3x} = 3^{6x} = y^6$	M1 A1 **(2)**
		(5 marks)

3.(a)	Attempts $\left(\frac{dy}{dx} =\right) 2x+3$ at $x=3$	M1
	At $x=3$ gradient of tangent = 9	A1 **(2)**
(b)	$\left(y_Q =\right) (3+h)^2+3(3+h)-2$	B1
	Gradient $PQ = \frac{(3+h)^2+3(3+h)-2-16}{3+h-3} = \frac{9h+h^2}{h} = 9+h$	M1 A1 **(3)**
(c)	States as $h \to 0$ Gradient $PQ \to 9$ = Gradient of tangent	B1 **(1)**
		(6 marks)

4 (a)	States or uses $\cos AOD = \frac{4}{12} \Rightarrow$ angle $AOD = 1.231$ *	M1 A1*
		(2)
(b)	Attempts $\frac{1}{2}r^2\theta$ with $r = 12$ and $\theta = \pi \pm 1.231$ or 1.231	M1
	Attempts area $AOD = \frac{1}{2} \times "4" \times \sqrt{12^2 - "4"^2}$ oe (22.627...)	M1
	Attempts sector − triangle $= \frac{1}{2} \times 12^2 \times (\pi + 1.231) - \frac{1}{2} \times "4" \times \sqrt{12^2 - "4"^2}$ (314.8….) – (22.627…)	ddM1
	or Attempts circle-sector-triangle $\pi \times 12^2 - \frac{1}{2} \times 12^2 \times (\pi - 1.231) - \frac{1}{2} \times "4" \times \sqrt{12^2 - "4"^2}$ 452.38… – 137.562…. – 22.627…… = awrt 292.2 (m^2)	A1
		(4)
(c)	Attempts $s = r\theta$ with $r = 12$ and $\theta = \pi \pm 1.231$ or 1.231	M1
	Attempts $P = 16 + \sqrt{12^2 - 4^2} + 12(\pi + 1.231)$ oe	dM1
	= awrt 79.8 (m)	A1
		(3)
		(9 marks)
Alt(a)	$AD = \sqrt{12^2 - 4^2} = 8\sqrt{2}$ $\cos AOD = \frac{12^2 + 4^2 - (8\sqrt{2})^2}{2 \times 12 \times 4} \Rightarrow$ angle $AOD = \cos^{-1}\left(\frac{1}{3}\right) = 1.231$	M1A1*

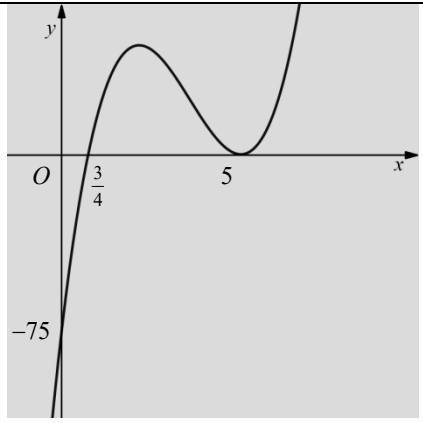
5. (a)	$20x^3 - 50x^2 - 30x = 0 \Rightarrow 10x\left(2x^2 - 5x - 3\right) = 0$	
	$\Rightarrow 10x\left(2x+1\right)\left(x-3\right) = 0$	M1
	$\Rightarrow x = 0, -\frac{1}{2}, 3$	A1, A1
		(3)
(b)	Sets or implies $\left(y+3\right)^{\frac{1}{2}} = 0$ or $-\frac{1}{2}$ or 3	B1ft
	Full method to find y	M1
	$y = 6$	A1ft,
	$y = -3,\ 6$	A1
		(4)
		(7 marks)

6.(a)	Attempts to find the gradient of $3x-4y+20=0 \Rightarrow y=\underline{\underline{\frac{3}{4}}}x+5$	M1
	Equation l_2 is $y-0="\frac{3}{4}"(x-8) \Rightarrow 3x-4y-24=0$ oe	M1, A1
		(3)
(b)	$P=\left(-\frac{20}{3},0\right), Q=(0,5)$	B1
	Area $PQRS=$ $PR\times OQ=\left(8+"\frac{20}{3}"\right)\times"5"=\frac{220}{3}$	M1, A1
		(3)
(c)	$\overrightarrow{QR}=\begin{pmatrix}8\\-5\end{pmatrix} \Rightarrow S=\left(-\frac{20}{3}+8,0-5\right)=\left(\frac{4}{3},-5\right)$	M1, A1
		(2)
		(8 marks)
Alt(c)	Solve their $y="\frac{3}{4}x-6"$ with their $"y=-\frac{5}{8}\left(x+\frac{20}{3}\right)"$	
	$\frac{3}{4}x-6=-\frac{5}{8}\left(x+\frac{20}{3}\right) \Rightarrow \frac{11}{8}x=\frac{11}{6} \Rightarrow x=...$	M1
	$S=\left(\frac{4}{3},-5\right)$	A1
		(2)

7. (a)(i)	$P = (90°, 3)$	M1, A1
(ii)	$Q = (540°, 0)$	B1 **(3)**
(b)	$(270°, 4)$	M1, A1 **(2)** **(5 marks)**

8.	Equates $y = k(2x-1)$ and $y = x^2 + 2x + 11 \Rightarrow k(2x-1) = x^2 + 2x + 11$	M1
	$\Rightarrow x^2 + (2-2k)x + 11 + k \quad (=0)$	A1
	Attempts "$b^2 - 4ac$"...0 $\Rightarrow (2-2k)^2 - 4(11+k)...0$	
	and proceeds to critical values	M1
	Critical values of $(k=) \quad 5, -2$	A1
	No roots so $b^2 - 4ac < 0$ so choose inside region	M1
	$-2 < k < 5$	A1
		(6)
		(6 marks)

9.	$\frac{4x^2+9}{2\sqrt{x}} = \frac{4x^2}{2\sqrt{x}} + \frac{9}{2\sqrt{x}} = 2x^{\frac{3}{2}} + \frac{9}{2}x^{-\frac{1}{2}}$	M1 A1
	$\left(\frac{dy}{dx} =\right) \quad 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}}$	M1 A1
	$\left(\frac{dy}{dx} =\right) \quad 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}} = 0 \Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \frac{\sqrt{3}}{2}$	M1 A1
		(6) **(6 marks)**
Alt(I)	Quotient rule	
	$u = 4x^2 + 9, u' = 8x, v = 2\sqrt{x}, v' = x^{-\frac{1}{2}}$	M1A1
	$\left(\frac{dy}{dx} =\right) \quad \frac{2\sqrt{x} \times 8x - (4x^2+9) \times x^{-\frac{1}{2}}}{4x}$	M1A1
	$\left(\frac{dy}{dx} =\right) \quad \frac{2\sqrt{x} \times 8x - (4x^2+9) \times x^{-\frac{1}{2}}}{4x} = 0 \Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \frac{\sqrt{3}}{2}$	M1A1
Alt(II)	Product rule	
	$u = 4x^2 + 9, u' = 8x, v = \frac{1}{2}x^{-\frac{1}{2}}, v' = \frac{1}{4}x^{-\frac{3}{2}}$	M1A1
	$\left(\frac{dy}{dx} =\right) \quad (4x^2+9) \times \frac{1}{4}x^{-\frac{3}{2}} + 8x \times \frac{1}{2}x^{-\frac{1}{2}}$	M1A1
	$\left(\frac{dy}{dx} =\right) \quad \frac{2\sqrt{x} \times 8x - (4x^2+9) \times x^{-\frac{1}{2}}}{4x} = 0 \Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \frac{\sqrt{3}}{2}$	M1A1

10.(a)		Shape for +ve x^3 Cuts x-axis at $\left(\frac{3}{4},0\right)$ and meets at $(5,0)$ Crosses y-axis at $(0,-75)$	B1 B1 B1 **(3)**
(b)(i)	$(x=)$ 3, 20		B1ft
(ii)	$(p=)75$		B1ft **(2)**
(c) (i)	$(g(x)=)\quad (4(x+1)-3)(x+1-5)^2=(4x+1)(x-4)^2$		M1 A1
(ii)	16		B1 **(3)** **(8 marks)**

11	**(a)**	Gradient of normal $= \frac{1}{4}$	B1
		Equation of normal $(y+50) = "\frac{1}{4}"(x-4) \Rightarrow y = \frac{1}{4}x - 51$	M1 A1
			(3)
	(b)	$\left(f''(x) =\right) \frac{6}{\sqrt{x^3}} + x = 6x^{-\frac{3}{2}} + x \Rightarrow f'(x) = -12x^{-\frac{1}{2}} + \frac{1}{2}x^2 + k$	M1 A1
		Substitutes $x = 4,\ f'(x) = -4 \Rightarrow k = -6$	dM1 A1
		$\left(f'(x) =\right) -12x^{-\frac{1}{2}} + \frac{1}{2}x^2 - 6 \Rightarrow \left(f(x) =\right) -24x^{\frac{1}{2}} + \frac{1}{6}x^3 - 6x + d$	dM1 A1ft
		Substitutes $x = 4,\ f(x) = -50 \Rightarrow d = \frac{34}{3}$	dddM1
		$\left(f(x) =\right) -24x^{\frac{1}{2}} + \frac{1}{6}x^3 - 6x + \frac{34}{3}$	A1
			(8)
			(11 marks)